

Clinical Fall Risk Summary for Patient ID: 3530

Model Prediction

Fall Classification: Fall
Fall Severity Classification: Recurrent Fall

Factors are ordered from most to least influential based on their contribution to the model prediction.

Factors Pushing the Prediction Toward Higher Severity (Recurrent Fall)

#	Factor	Patient Value	Interpretation / Scale
1	UPDRS-III Total	56	MDS-UPDRS Part III total motor examination score (sum of 33 items). 0-132; higher = more severe motor impairment
2	MoCA	15	Montreal Cognitive Assessment (MoCA) total cognition score. 0-30; higher = better cognition
3	Mobility Summary	3.89	Gaussian-weighted mobility summary across seven daily activities (toilet transfers, curbs, car transfers, bed-to-chair, errands/shopping, getting off the floor, walking 15+ minutes). Higher values indicate more responses in the moderate-difficulty range; this score is non-monotonic and does not simply mean greater overall mobility difficulty
4	PD Duration	15	Years since Parkinson's disease diagnosis. Non-negative integer years

Factors Pushing the Prediction Toward Lower Severity (Rare Fall)

#	Factor	Patient Value	Interpretation / Scale
1	FOG (Freezing of Gait)	1	Self-reported severity of freezing of gait. 0 = None; 1 = Rare freezing (may have start hesitation); 2 = Occasional freezing when walking; 3 = Frequent freezing with occasional falls; 4 = Frequent falls from freezing
2	BMI	43.04	Body mass index. kg/m^2
3	Urinary Problems	3	Urinary problems (MDS-UPDRS Part I). 0 = Normal (no urine control problems); 1 = Slight (urinate often/urgently no daily activity impact); 2 = Mild (some difficulties with daily activities no accidents); 3 = Moderate (a lot of difficulties including urine accidents); 4 = Severe (cannot control urine use protective garment or bladder tube)
4	UPDRS-IV Total	5	MDS-UPDRS Part IV total motor complications score (dyskinesia and motor fluctuations). 0-24; higher = more motor complications
5	Gait	1	Clinician-rated gait impairment (MDS-UPDRS Part III). 0 = Normal (no problems); 1 = Slight (independent walking with minor impairment); 2 = Mild (independent walking with substantial impairment); 3 = Moderate (requires walking aid but not a person); 4 = Severe (cannot walk without another person's assistance); 101 = unable to rate
6	Postural Instability at Dx	0	Postural instability present at diagnosis. 0 = No; 1 = Yes; 2 = Unknown

Model Interpretation

Within this model, the Recurrent Fall classification was primarily associated with the patient's motor impairment, cognitive status, and disease duration. These factors, alongside the mobility summary score, were the primary drivers for the model's output. Conversely, the presence of rare freezing of gait and the patient's BMI were associated with the lower severity classification, though these did not outweigh the factors supporting the Recurrent Fall prediction.

Clinical Note

- Model predictions are intended to support clinical review and should not replace clinical judgment.
- The factors shown above identify the patient characteristics that most influenced the model prediction and provide insight into the model's reasoning.
- For definitions and interpretation of all variables used by the model, please refer to the Feature Reference Guide (feature_map.xlsx).